

After Hrushovski, after Simon

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ABSTRACT.

1. Patterns

In the following $\mathcal{U}$ is some monster model of a complete theory T of language L . For simplicity T is assumed to be one-sorted.

We define a structure whose domain is a subset of $\mathcal{U}^x$, where x is some fixed tuple of variables. The language is denoted by L^P . Though L^P depends on x , this will be left implicit.

Let V be some large set of variables that are copies of x . Let Δ be the set of positive Boolean combinations of formulas in $\bigcup_{x' \in V} L_{x'}; z$, where $z = \langle z_i : i < \lambda \rangle$ is a fixed tuple of variables. The choice of Δ is justified in Remark 1.

For every formula $\varphi(x_0, \dots, x_{n-1}; z) \in \Delta$ and every type $\alpha(z) \in L$ the language L^P contains the n -ary relation $r_{\varphi; \alpha}$.

Below by the symbol L^P we will also denote the set of *atomic* formulas $r_{\varphi; \alpha}(x_0, \dots, x_{n-1})$ for any $x_0, \dots, x_{n-1} \in V$. Note that equalities are *not* included in L^P .

For a given pairs of models $M \leq N$, where N is $|M|^+$ -saturated, we define a model $\mathcal{N}$ with domain N^x . We will show that $\mathcal{N}$ is very saturated (in a suitable language) but, as N is taken to be small w.r.t. $\mathcal{U}$, we will be able to realize in $\mathcal{U}$ all types in $S(\mathcal{N})$.

We define the interpretation of $r_{\varphi; \alpha}$ in $\mathcal{N}$. If $\bar{a} = a_0, \dots, a_{n-1} \in \mathcal{N}$, we declare $r_{\varphi; \alpha}(\bar{a})$ true if $\varphi(\bar{a}; b)$ holds for every $b \in M^z$ such that $\alpha(b)$. We repeat the definition for emphasis

$$r_{\varphi; \alpha}(\bar{a}) \Leftrightarrow \alpha(M^z) \subseteq \varphi(\bar{a}; M^z).$$

The following is paramount.

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1 Remark By the choice of Δ , the truth of $r_{\varphi;\alpha}(\bar{a})$ only depends on the equivalence class of $\equiv_M$ of the components of the tuple $\bar{a}$. Explicitely, if $\bar{a} = a_0, \dots, a_{n-1}$ and $\bar{a}' = a'_0, \dots, a'_{n-1}$ are such that $a_i \equiv_M a'_i$ for every $i < n$, then

$$\# \quad r_{\varphi;\alpha}(\bar{a}) \leftrightarrow r_{\varphi;\alpha}(\bar{a}').$$

But we can stretch this further. Write $a \equiv_{\{M\}} a'$ if $f a = a'$ for some $f \in \text{Aut}(N/\{M\})$ i.e. a and a' are conjugated by an automorphism that fixes M setwise. Then $\#$ holds whenever $a_i \equiv_{\{M\}} a'_i$ for every $i < n$.

By the remark, it would be natural to define the interpretation of L^P on the quotient $\mathcal{N}/\equiv_M$ or even on $\mathcal{N}/\equiv_{\{M\}}$ – albeit the latter is more perplexing and will not be considered here. But we prefer to keep the original domain $\mathcal{N}$ to have a common playground with L .

A **p-type** is a set of (atomic) formulas in $L^P(\mathcal{N})$.

Note incidentally that the relation $x_0 \equiv_M x_1$ is definable by a p-type. Namely the p-type containing the formulas $r_{\vartheta;\top}(x_0, x_1)$ for every $\vartheta(x_0, x_1; z)$ of the form $\varphi(x_0; z) \leftrightarrow \varphi(x_1; z)$.

We write **p-tp**($\bar{a}$) for the p-type of $\bar{a}$, i.e. the set of formulas $r_{\varphi;\alpha}(\bar{x})$ such that $r_{\varphi;\alpha}(\bar{a})$ holds.

2 Fact Formulas in L^P are closed under adjunction of dummy variables, disjunction, conjunction, and existential quantification.

Proof. The claim about dummy variables is clear, so we tacitly use it below.

Up to renaming of the variables that occur in $\beta(z)$ and $\psi(\bar{x}; z)$, we can assume that they are disjoint from those in $\alpha(z)$. Therefore the following equivalences hold

$$\begin{aligned} r_{\varphi;\alpha}(\bar{a}) \vee r_{\psi;\beta}(\bar{a}) &\Leftrightarrow \forall z \in \alpha(M^z) \varphi(\bar{a}; z) \vee \forall z \in \beta(M^z) \psi(\bar{a}; z) \\ &\Leftrightarrow \forall z \in \alpha(M^z) \cap \beta(M^z) \varphi(\bar{a}; z) \vee \psi(\bar{a}; z) \\ &\Leftrightarrow r_{\varphi \vee \psi; \alpha \cap \beta}(\bar{a}) \end{aligned}$$

A similar argument applies to the conjunction of two formulas.

Finally, consider the formula $\exists y r_{\varphi;\alpha}(\bar{x}, y)$. Without loss of generality, we can assume that φ has the form

$$\varphi(\bar{x}, y; z) = \bigvee_{i=1}^m \varphi_i(\bar{x}, y; z)$$

where $\varphi_i(\bar{x}, y; z) = \varphi'_i(\bar{x}; z) \wedge \psi_i(y; z)$ and each $\varphi'_i(\bar{x}; z)$ is a conjunction of formulas in $\bigcup_{x' \in V} L_{x'; z}$. Closure under existential quantification follows from the following equivalences

$$\begin{aligned} \exists y r_{\varphi;\alpha}(\bar{a}, y) &\Leftrightarrow \exists y \forall z \left[z \in \alpha(M) \rightarrow \bigvee_{i=1}^m \varphi_i(\bar{a}, y; z) \right] \\ &\Leftrightarrow \forall z \left[z \in \alpha(M) \rightarrow \bigvee_{i=1}^m \varphi'_i(\bar{a}; z) \wedge \exists y \psi_i(y; z) \right] \end{aligned}$$

$$\Leftrightarrow r_{\vartheta;\alpha}(\bar{a}),$$

where

$$\vartheta(\bar{x};z) = \bigwedge_{i=1}^m \varphi'_i(\bar{x};z) \wedge \exists y \psi_i(y;z)$$

is a positive Boolean combination of formulas in $\bigcup_{x' \in V} L_{x';z}$, as required. $\square$

3 Fact (∞ -p-saturation) Let $A \subseteq \mathcal{N}$ have arbitrary cardinality. Let $\bar{x}$ be a tuple of copies of x of length $\leq |N|$. Then every p-type $p(\bar{x}) \subseteq L^p(A)$ that is finitely consistent in $\mathcal{N}$ is realized in $\mathcal{N}$.

Proof. Let $A \subseteq \mathcal{U}^x$ be a small set containing a representative of every $\equiv_M$ equivalence class that intersects A . Replacing A in $p(\bar{x})$ by elements A we obtain a equivalent type $p'(\bar{x}) \subseteq L^p(A)$. Note that every formula $r_{\varphi;\alpha}(\bar{x})$ there is a type $q(\bar{x}) \subseteq L$ such that

$$\mathcal{N} \models r_{\varphi;\alpha}(\bar{a}) \Leftrightarrow \mathcal{N} \models q(\bar{a})$$

Therefore the saturation of N implies that $p'(\bar{x})$ is realized in $\mathcal{N}$. $\square$

2. Morphism

As the interpretation of L^p factors through the equivalence relation $\equiv_M$, p-morphisms could be introduced as (partial) maps of $\mathcal{U}^x/\equiv_M$ into itself. However there are a few occasions when it is preferable to work with maps of $\mathcal{U}^x$ into itself. Though the switching between the two perspectives is intuitively obvious, it requires a fair amount of notation.

Let $R \subseteq \mathcal{U}^x \times \mathcal{U}^x$ be a binary relation. If $\bar{a} = \langle a_i : i < \lambda \rangle$ and $\bar{b} = \langle b_i : i < \lambda \rangle$ we write $\langle \bar{a}, \bar{b} \rangle \in R$ to abbreviate: $\langle a_i, b_i \rangle \in R$ for all $i < \lambda$. We say that R is a **p-morphism** if the following condition holds

$$\# \quad \mathcal{U}^x \models r_{\varphi;\alpha}(\bar{a}) \Rightarrow \mathcal{U}^x \models r_{\varphi;\alpha}(\bar{b}) \quad \text{for every } r_{\varphi;\alpha} \in L^p \text{ and every } \langle \bar{a}, \bar{b} \rangle \in R.$$

When the converse implication in $\#$ also holds, i.e. when also R^{-1} is a p-morphism, we say that R is a **partial p-isomorphism**. When $\mathcal{A} \subseteq \text{dom } R$, $\mathcal{B} \subseteq \text{range } R$, and $R \cap (\mathcal{A} \times \mathcal{B})$ is a partial p-isomorphism, we say that R is an **p-isomorphism** between $\mathcal{A}$ and $\mathcal{B}$ or, when $\mathcal{A} = \mathcal{B}$, a **p-automorphism** of $\mathcal{A}$.

If R is a p-morphism, also $\equiv_M \circ R \circ \equiv_M$ is a p-morphism. This we call the **full** p-morphism generated by R . We say that R and R' are **equivalent** if they generate the same full p-morphism.

4 Remark We can identify p-morphisms with maps $f : \mathcal{U}^x / \equiv_M \rightarrow \mathcal{U}^x / \equiv_M$. In fact, to any such map we can associate the relation $R = \bigcup_{a \in \mathcal{U}^x} ((a / \equiv_M) \times f(a / \equiv_M))$.

Vice versa, to any p-morphisms R we can associate $f : \mathcal{U}^x / \equiv_M \rightarrow \mathcal{U}^x / \equiv_M$ that maps $a / \equiv_M \mapsto b / \equiv_M$ for some/any b such that $\langle a, b \rangle \in R$. This is well-defined because if $\langle a, b \rangle, \langle a', b' \rangle \in R$ and $a \equiv_M a'$, then we necessarily have $b \equiv_M b'$.

We find more convenient to work with relations.

We say that the p-morphism R is **reduced** if there is no $R' \subset R$ that is equivalent to R . It is clear that if $A, B \subseteq \mathcal{U}^x$ are any two sets of representative of $\equiv_M$ then $f = R \cap (A \times B)$ is a reduced morphism equivalent to R .

5 Fact Any reduced p-morphism is the graph of a (partial) map. Therefore every reduced p-isomorphism is a bijection.

Proof. Suppose not, then there are a, b, c such that $\langle a, b \rangle, \langle a, c \rangle \in R$ but $b \neq c$. As R is reduced, then $b \not\equiv_M c$. As noted above, this is not possible. The second claim follows immediately. $\square$

6 Fact Every p-morphism extends to a p-morphism defined on the whole of $\mathcal{U}^x$. Also, every p-isomorphism extends to a p-automorphism of $\mathcal{U}^x$.

Proof. Let R be a p-morphism. Pick be a reduced p-morphism f equivalent to R . Let $\bar{a}$ be an enumeration of $\text{dom } f$ and let $b \in \mathcal{U}^x$. Let $p(\bar{x}) = \text{p-tp}(\bar{a})$ and let $q(\bar{x}, y) = \text{p-tp}(\bar{a}, b)$. By ∞ -p-saturation $\exists y q(\bar{x}, y)$ is equivalent to the type containing the existential closure of the formulas in $q(\bar{x}, y)$. By Fact 2, the latter is equivalent to a p-type, which is necessarily coincides with $p(\bar{x})$. Then $q(f\bar{a}, y)$ is consistent hence by ∞ -p-saturation it is realized by some $c \in \mathcal{U}^x$. Then $f \cup \{\langle b, c \rangle\}$ is a p-morphism defined in b that extends f . Then R can be extended as well.

The second claim is obtained by back-and-forth. $\square$

3. Duality

In this section $\bar{m}$ is an enumeration of M and $|z| = |\bar{m}|$. For convenience, x is taken of infinite length. Therefore we can assume that the formulas in the types in $S_z(\mathcal{U})$ are all of the form $\varphi(a; z)$ for some $\varphi(x; z) \in L$ and $a \in \mathcal{U}^x$.

7 Remark We remark a simple fact which is used in the proof below. We say that $\varphi'(a; z)$ is an alphabetic variant of $\varphi(a; z)$ if it is obtained by replacing the variables effectively occurring in $\varphi(a; z)$ by other variables among those in z . The following are clearly equivalent

1. $\varphi(a; M^z) = M^z$
2. $\varphi'(a; \bar{m})$ holds for every alphabetic variant of $\varphi(a; z)$.

Let $\mathcal{B} = \{b \in \mathcal{U}^z : \bar{m} \equiv b \perp_M N\}$.

8 Fact Then there is a one-to-one correspondence between the elements of $\mathcal{B}/\equiv_M$ and the total p-morphisms of $\mathcal{N}$.

Proof. In 1 below, given a p-morphism, we define an element of $\mathcal{B}$. In 2, given an element of $\mathcal{B}$, we construct a p-morphism.

1. Let $f : \mathcal{N}/\equiv_M \rightarrow \mathcal{N}/\equiv_M$ be a total p-morphism. Let $p(z) \subseteq L(N)$ be the type containing the formulas $\varphi(a; z)$ such that $\varphi(fa; \bar{m})$ for every $a \in \mathcal{N}$. Clearly $p(z)$ is complete, and contains $\text{tp}(\bar{m})$. We prove that $p(z)$ is finitely satisfied in M , then the fact is proved if we define b to be some/any realization of $p(z)$.

Suppose for a contradiction that $p(z)$ is not finitely satisfied in M . Then $\psi(a_1, \dots, a_n; M^z) = \emptyset$, where

$$\psi(x_1, \dots, x_n; z) = \bigwedge_{i=1}^n \varphi_i(x_i; z)$$

for some $\varphi_i(x; z) \in L$ and $a_i \in \mathcal{N}$ such that $\varphi_i(fa_i; \bar{m})$ for every i . Then $r_{\neg\psi; \top}(a_1, \dots, a_n)$ and $\neg r_{\neg\psi; \top}(fa_1, \dots, fa_n)$. This is not possible because f is a p-morphism.

2. Let $b \in \mathcal{B}$ be given – the reader may check that the construction only depends on $b/\equiv_M$. For every $a \in \mathcal{N}$ we define $p_a(x) = \{\varphi(x, \bar{m}) : \varphi(a; b)\}$. We claim that if $a \equiv_M a'$ then $p_a(x) = p_{a'}(x)$. In fact, $\varphi(a; b) \leftrightarrow \varphi(a'; b)$ would contradict $b \perp_M N$. We also claim that $p_a(x)$ is consistent, otherwise $\neg \exists x \bigwedge_i \varphi_i(x; \bar{m})$, for some finitely many formulas $\varphi_i(x; z)$. But this would contradict $b \equiv \bar{m}$.

Then we can define $f(a)$ to be some/any realization of $p_a(x)$.

We prove that f is a p-morphism. Let $\psi(x_1, \dots, x_n; z) \in \Delta$, $\alpha(z) \in L$, and $a_1, \dots, a_n \in \mathcal{U}^x$ be such that $r_{\psi; \alpha}(a_1, \dots, a_n)$. Assume

$$\psi(x_1, \dots, x_n; z) = \bigvee_{h=1}^m \bigwedge_{k=1}^m \varphi_{h,k}(x_{i_{h,k}}; z)$$

for some $i_{h,k} \in \{1, \dots, n\}$. Now, after re-indexing, from $r_{\psi; \alpha}(a_1, \dots, a_n)$ we obtain that $\alpha(M^z) \subseteq \varphi_k(a_k; M^z)$ for $k = 1, \dots, m$. Then from $b \perp_M N$ we obtain that b satisfies $\alpha(z) \rightarrow \varphi_k(a_k; z)$ and all its alphabetic variants. Therefore $\bar{m}$ satisfies $\alpha(z) \rightarrow \varphi_k(fa_k; z)$ and all its alphabetic variants. Therefore $\alpha(M^z) \subseteq \varphi_k(fa_k; M^z)$ and $r_{\psi; \alpha}f(a_1, \dots, fa_n)$ follows. $\square$

4. Semigroups

Assume $\mathcal{U}$ is a semigroup w.r.t. some definable operation which we denote by $\cdot$. In this section $|x| = 1$ and, as always, $z = \langle z_i : i < \lambda \rangle$. The set Δ contains Boolean combinations of formulas of the form $\varphi(z_i \cdot x')$ for $\varphi(x) \in L$, and x' any copy of x .

9 Fact For every $b \in \mathcal{B}$ the map $f : \text{tp}(a/M) \mapsto \text{tp}(b \cdot a/M)$ is a well-defined p-morphism.

Proof. We prove that f is well-defined. Pick any two $a \equiv_M a'$ in $\mathcal{N}$. As $b \perp_M N$ then $a \equiv_{M,b} a'$, it follows that $b \cdot a \equiv_M b \cdot a'$ as required.

Now, assume $r_{\psi,\alpha}(a_1, \dots, a_n)$. We prove $r_{\psi,\alpha}(fa_1, \dots, fa_n)$. Assume that

$$\psi(x_1, \dots, x_n; z) = \bigvee_{h=1}^m \bigwedge_{k=1}^m \varphi_{h,k}(z_{i_{h,k}} \cdot x_{j_{h,k}})$$

where $i_{h,k} < \lambda$ and $j_{h,k} \in \{1, \dots, n\}$. From $r_{\psi,\alpha}(a_1, \dots, a_n)$, after re-indexing, we obtain that $\alpha(M^z) \subseteq \varphi_k(M^z \cdot a_k)$ for $k = 1, \dots, m$. As $b \perp_M N$, we immediately obtain $\alpha(M^z) \subseteq \varphi_k(M^z \cdot b \cdot a_k)$ for all k , so $r_{\psi,\alpha}(fa_1, \dots, fa_n)$ follows. $\square$

5. Cores

A set $A \subseteq \mathcal{N}$ is **p-maximal** if it satisfies the following (obviously) equivalent conditions for every $\bar{a} \in A$ and every p-morphism f

- a. $r_{\varphi;\alpha}(\bar{a}) \Leftrightarrow r_{\varphi;\alpha}(f\bar{a})$ for every $r_{\varphi;\alpha} \in L^p$
- b. $\text{p-tp}(\bar{a}) = \text{p-tp}(f\bar{a})$.

10 Fact The following are equivalent

1. $A \subseteq \mathcal{N}$ is p-maximal
2. $p(\bar{x}) = \text{p-tp}(\bar{a})$ is a maximally consistent p-type for every $\bar{a} \in A^{\bar{x}}$ and every finite $\bar{x}$
3. $p(\bar{y}) = \text{p-tp}(\bar{c})$ is a maximally consistent p-type for $\bar{c}$ any enumeration of A
4. every p-morphism defined on A is a p-isomorphism between A and its image.

Proof. 1 $\Rightarrow$ 2. Negate 2. Suppose $\text{p-tp}(\bar{a})$ is not maximal for some finite $\bar{a} \in A^{\bar{x}}$. Pick some $\bar{a}' \in A^{\bar{x}}$ such that $\text{p-tp}(\bar{a}) \subset \text{p-tp}(\bar{a}')$. Then $f = \{\langle \bar{a}, \bar{a}' \rangle\}$ is a p-morphism that contradicts the p-maximality of A .

2 $\Rightarrow$ 3. If $\bar{c}$ is an enumeration of A that witnesses the failure of 3, then some finite subtuple of $\bar{c}$ witnesses the failure of 2.

3 $\Rightarrow$ 4. Negate 4. Let f be a p-morphism defined on A that is not a p-isomorphism. Let $\bar{a} \in A^{\bar{x}}$ and $\bar{b}$ be such that $\langle \bar{a}, \bar{b} \rangle \in R$ and $\text{p-tp}(\bar{a}) \sim \text{p-tp}(\bar{b}) \neq \emptyset$. To contradict 3, it suffices to extend $\{\langle \bar{a}, \bar{b} \rangle\}$ to a p-morphism defined on A . This is possible by ∞ -p-homogeneity.

4 $\Rightarrow$ 1. Clear. $\square$

11 Example Assume that $a \in \mathcal{U}^x$ is such that $\varphi(a; M^z)$ is definable over $\text{acl}^{\text{eq}} \emptyset$ for every $\varphi(x; z) \in L$. Then the orbit of a under $\text{Aut}(\mathcal{U}/\{M\})$ is p-maximal.

Proof. First consider the case when $\varphi(a; M^z)$ is definable over $\emptyset$. We prove that $\{a\}$ is p-maximal.

Let $\varphi'(z) \in L$ be such that $\varphi(a; M^z) \leftrightarrow \varphi'(M^z)$. Then, if we set $\varphi''(x; z) = \varphi(x; z) \leftrightarrow \varphi'(z)$, the formula $r_{\varphi''; \top}(a)$ holds.

If $\{\langle a, b \rangle\}$ is a p-morphism then $r_{\varphi''; \top}(b)$ also holds. Then $a \equiv_M b$ follows. This shows that p-morphisms are trivial and therefore p-isomorphisms.

Now assume that $\varphi(a; M^Z)$ is definable over $\text{acl}^{\text{eq}} \emptyset$.???

□

A set $\mathcal{J} \subseteq \mathcal{U}^x$ is called a **core** if it is p-maximal and, among the p-maximal sets, it is maximal w.r.t. inclusion.

12 Fact Let $\mathcal{J}$ be a core. Let $\mathcal{A}$ be a p-maximal set. Then every p-morphism $R \subseteq \mathcal{A} \times \mathcal{J}$ extends to a p-morphism R' such that $\mathcal{A} = \text{dom} R'$ and $\text{range} R' \subseteq \mathcal{J}$.

Proof. Reason as in the proof of Fact 6 with the same notation but choose $b \in \mathcal{A}$. Let R' be a p-morphism defined on $\mathcal{A}$ that extends $f \cup \{\langle b, c \rangle\}$. Then $R'[\mathcal{A}]$, i.e. the image of $\mathcal{A}$ under R' , is a p-maximal set that contains $\mathcal{J}$. By maximality $R'[\mathcal{A}]$ coincides with $\mathcal{J}$, hence $c \in \mathcal{J}$. □

By back-and-forth we obtain the following.

13 Corollary (∞ -p-homogeneity of the core) If $\mathcal{J}$ and $\mathcal{J}'$ are cores, then every p-morphism $R \subseteq \mathcal{J} \times \mathcal{J}'$ extends to a p-isomorphism between $\mathcal{J}$ and $\mathcal{J}'$.

A **p-retraction** of $\mathcal{U}^x$ onto $\mathcal{A}$ is a total p-morphism R that coincides with $\equiv_M$ on $\mathcal{A}$, i.e. such that $R \cap (\mathcal{A} \times \mathcal{A}) = (\equiv_M) \cap (\mathcal{A} \times \mathcal{A})$.

14 Theorem If $\mathcal{J}$ is a core then there is a p-retraction of $\mathcal{U}^x$ onto $\mathcal{J}$.

Proof. Let $\bar{c}$ enumerate $\mathcal{U}^x$. Let $\bar{a}$ be a tuple such that $\text{p-tp}(\bar{a}) \supseteq \text{p-tp}(\bar{c})$ is p-maximal. By Fact ?? the relation $R = (\equiv_M \circ \{\langle \bar{c}, \bar{a} \rangle\} \circ \equiv_M)$ is a p-morphism onto the p-maximal set $\mathcal{A} = \text{range } R$. As $\mathcal{A}$ is p-maximal, by Fact 12 we can extend $(R \cap (\mathcal{J} \times \mathcal{A}))^{-1}$ to a p-isomorphism R' defined on $\mathcal{A}$. It is easy to verify that $R' \circ R$ is a retraction. □

15 Fact The following are equivalent

1. $a \in \bigcap \{\mathcal{J} : \mathcal{J} \text{ is a core}\}$
2. $\varphi(a; M^Z)$ is definable over $\text{acl}^{\text{eq}} \emptyset$, for every formula $\varphi(x; z) \in L$.

Proof. ???

□

16 Definition Let $\mathcal{J}$ be a core; we define a topology on $\mathcal{J}$. The closed sets in the p-topology on $\mathcal{J}$ are those defined by p-types with (arbitrary many) parameters in $\mathcal{J}$.

17 Fact The p-topology is compact and T_1 .

Proof. ???

□

6. Esperimenti

Let M be a saturated model and let $G = \text{Aut}(M)$.

The group G acts on $\mathcal{U}^x / \equiv_M$ in the natural way.